Day 4 Handout: Uncertainty, Bias, Limitations and the Capstone

Experimentation and Causal Inference, SDAIA Academy. Revision notes for participants.

What Day 4 was about

A causal estimate carries two kinds of doubt: sampling uncertainty (what standard errors capture) and identification uncertainty (what sensitivity analysis captures). Day 4 covered how to get the first right, how to quantify the second, how to name the bias you are fighting, and how to turn all of it into a decision whose risk the reader can see and own.

Uncertainty quantification

Two kinds of uncertainty. Sampling: would the estimate change under a different sample or re-randomization? Identification: would it change if the assumptions were slightly wrong? For observational work the second usually dominates. A tight confidence interval around a poorly identified estimate is false precision. Report both.

Choosing the standard error:

Type	When	Note
Classical (homoskedastic)	Rarely defensible	Usually too optimistic
Robust (HC2, HC3)	Cross-sectional regression, user-level experiments	The sensible default
Clustered	Grouped assignment or correlated units	Cluster at the level of randomization or the treatment shock, never finer
Design-based	Experiments	Derived from the assignment mechanism

Clustering is the most common uncertainty error in applied causal work. Treating correlated observations as independent (sessions within a user, people within a region) shrinks the SE several-fold and manufactures significance. With few clusters (13 regions), standard asymptotics fail; use a wild-cluster bootstrap or randomization inference.

Bootstrap: resample at the unit of independence (block bootstrap for time series, cluster bootstrap for grouped data). The naive bootstrap can fail for matching estimators and extreme quantiles.

Confidence intervals: a 95 percent interval is a property of the procedure (95 percent of such intervals contain the truth), not a probability statement about this one. A wide interval spanning zero says “underpowered or uncertain,” not “no effect.” Read it as the range of effects compatible with the data and model.

Practitioner checklist: SE matches the mechanism; clustering at the right level with enough clusters; interval reported, not just a point and p; sampling separated from identification uncertainty; interval translated into expected value and downside risk for the decision.

Bias, validity and sensitivity

Taxonomy of bias. Confounding: a common cause of treatment and outcome; fix by design or valid adjustment. Selection: how units enter or stay in the sample relates to treatment and outcome, including collider conditioning and attrition. Measurement: systematic error in T, Y or X, proxies, or differential measurement across arms. Interference: one unit’s treatment changes another’s outcome (SUTVA). Name the bias precisely and you know where in the pipeline to fix it.

Internal versus external validity. Internal: is the estimate right within the study (threatened by confounding, attrition, non-compliance)? External: does it transfer to other populations, settings, times or treatment versions? Tightly controlled designs (RCTs, RDD) buy internal validity and can narrow external validity; an effect at a cutoff or for compliers can be exactly right and still not answer the population question.

Sampling limitations. Coverage: opt-in panels and platform data can exclude the people a decision must serve. Effect transport: if effects vary with covariates and the target population’s mix differs, the sample ATE will not transport. Post-stratify or reweight to the target and state the assumption that licenses it.

Sensitivity analysis for unmeasured confounding. Since ignorability is untestable, quantify how strong a hidden confounder would need to be, with both treatment and outcome, to explain the effect away. Tools: E-values (E = RR + sqrt(RR (RR minus 1)) for a risk ratio RR), Rosenbaum bounds, partial R-squared benchmarks (Cinelli-Hazlett) against observed covariates. Report the tipping point and benchmark it: “a confounder as strong as X would be needed, and none such is plausible.” An E-value near 1 means the finding is fragile.

Placebo tests and negative controls. Negative-control outcome (something the treatment cannot affect), negative-control exposure (a pseudo-treatment), placebo timing or placebo units. A detected effect where none can exist reveals residual bias.

Threats-to-validity review, to run on any causal claim before acting: confounding, selection and attrition, measurement, interference, external validity, robustness.

Decision-making under causal uncertainty

The analyst’s job ends at a defensible decision, not at an estimate. Combine the effect distribution with payoffs and costs to compute expected value and downside risk. When uncertainty is decision-relevant, quantify the value of a further experiment before running it. A good causal report lets a reader see exactly which assumption the recommendation rests on and how the decision would change if it failed.

Communicating a result to leadership: lead with the effect size and interval in business units, not a p-value; state practical significance against the threshold agreed in the design; convey assumptions as risks; give a clear recommendation (ship, scale, hold, redesign) and say what evidence would change it; use one decisive visual (the interval plot or the event study).

“We cannot conclude” is a legitimate deliverable when it is correct: say what was ruled out, what remains open, and the concrete next step. A confident wrong answer is a failure; an honest “not yet” is a service.

Cross-case lessons from the three case studies: the naive observational number is almost always wrong and usually too large; the estimand includes time (week-one and steady-state effects differ); when you cannot randomize, triangulate across designs with different assumptions; high-stakes decisions demand sensitivity analysis, not just a tight interval; the deliverable is a decision with owned risk.

Method selection in one paragraph

Can we randomize going forward? Run an A/B test. Did a treatment switch on at a known time for some units? Difference-in-differences or synthetic control. Is there as-good-as-random variation in a confounded treatment? Instrumental variables. Is there a threshold rule? Regression discontinuity. Only cross-sectional data with measured confounders? Doubly robust adjustment on a DAG-derived backdoor set with a sensitivity analysis. No measured way to close the backdoors? Declare the effect not identifiable and say what is missing.

Capstone brief

You are the experimentation lead for Injaz. Two linked decisions: whether to launch the AI form-autofill feature nationally (a randomized test has run: `injaz_autofill_ab.csv`, 70,000 users, ship threshold +1.5 points on completion with no guardrail regression) and whether to scale the regional reminder program (rolled out region by region without randomization: `injaz_regions_panel.csv`).

Deliverable: a runnable notebook plus a two-page decision memo containing the estimand, the DAG or identifying assumption, the design, the estimate with interval, diagnostics, a sensitivity analysis, and a recommendation with the key assumption and what would change the call. A five-minute defense follows.

Rubric: estimand and question 15; identification and design 25; estimation and inference 25; diagnostics and sensitivity 20; communication and decision 15. Pass at 70, distinction at 90.

Pitfalls that cost points: a naive comparison without naming the identifying assumption; adjusting for a post-treatment variable; a standard error that does not match the design; a confident recommendation with no sensitivity analysis; an estimand that quietly differs from the decision; SHIP on an interval that does not clear the threshold; “no significant effect” written as “no effect.”

A great deliverable is transparent (DAG, assumptions and code visible), diagnosed (overlap, balance and design checks shown, not asserted), honest about uncertainty (both layers, with a sensitivity bound), and actionable (ends in a decision with the risk explicitly owned by the reader).

Self-check (answers at the end)

1. A regional policy is assigned per region but you have person-level data. Cluster at what level?
2. A 95 percent CI of [minus 0.2, 3.4] on a lift means what?
3. An RDD estimate is unbiased at the cutoff but may not generalize. This is a limit of which validity?
4. A large E-value indicates what?
5. Detecting an “effect” on a negative-control outcome suggests what?
6. You have grouped assignment, thin overlap and a threshold rule. Most defensible design?
7. The single best summary of causal-inference craft?
8. When is “we cannot conclude” a full-marks answer?

Answers

1. Region.
2. The data cannot rule out a wide range, from a small negative to a large positive effect.
3. External validity.
4. The finding is robust to unmeasured confounding.
5. Residual confounding.
6. RDD at the threshold.
7. Design and identify before you estimate.
8. When it is correct and justified: the analysis states what was ruled out, what remains open, and the next step that would settle it.

The whole course in four lines

Every effect is a contrast of counterfactuals; the missing one is the problem to solve. Randomize when you can; when you cannot, borrow its logic and name your assumption. Match the estimator and the standard error to the design; quantify uncertainty at both layers. End in a recommendation whose risk the reader can see and own.